

Euclidean and Non-Euclidean Geometry:

Major Exercises:

1.

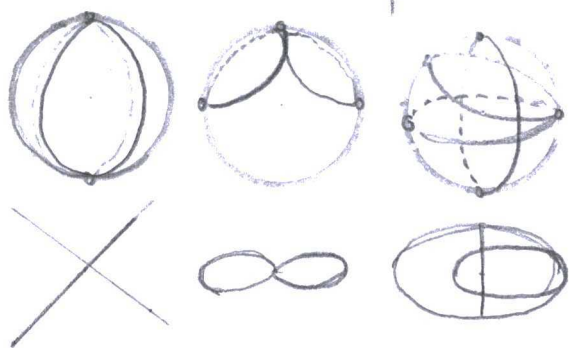
Citation: "Journey into Geometries"

Sved, Marta.

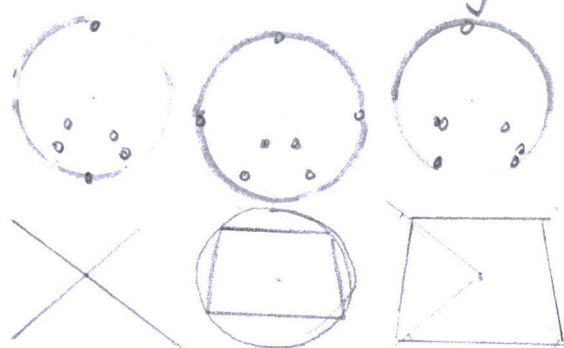
The mathematical association
of America (1991)

Chapter 3: "Dr. Whatif's Euclidean Geometry:"

Personal Interpretation:



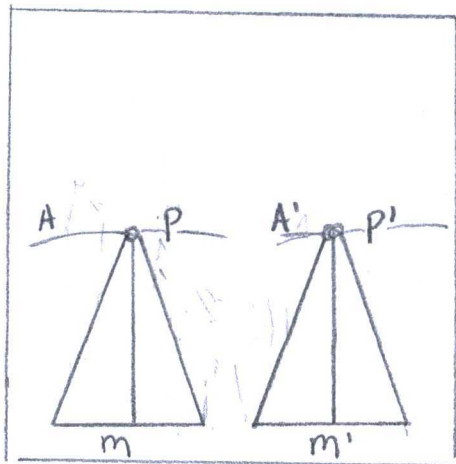
Punctured Geometry:



Parallel Property:

Euclidean	No	No	No		No	No	No
Elliptic	Yes	Yes	Yes		Yes	Yes	Yes
Hyperbolic	No	No	No		No	No	No
Isomorphic	No	No	No		Yes	No	No

2. Every Projective Plane is isomorphic to an affine plane:



If $P = \text{Projective plane}$
 $m = \text{line}$

then $P' = P - m$ is also a projective plane

If $A(P, m)$, then $A'(P', m')$

3.

a) Each point in P has $n+1$ lines through the projection:

By Definition as base case,
 projective planes follow:

1. Elliptic Parallel Property:

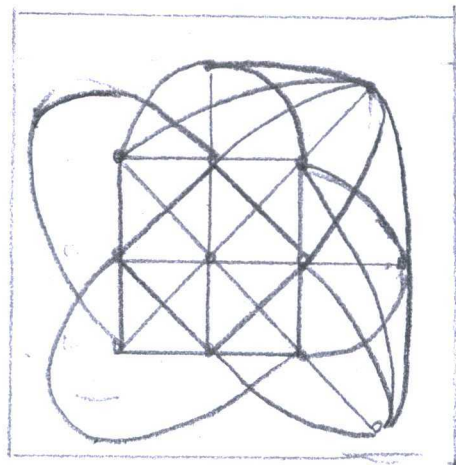
Two lines meet at a point.

2. Incidence Axiom #2

With two lines, one point
 incident with both of them.

b) Total number of points in P is $n^2 + n + 1$:

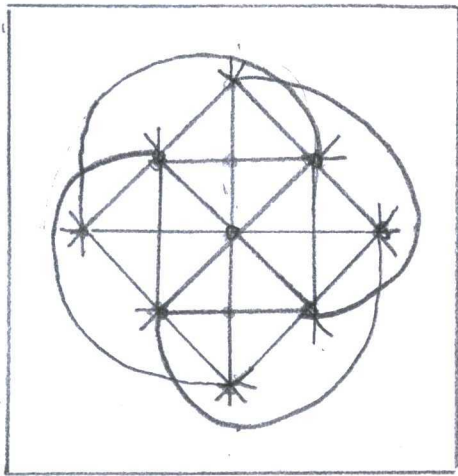
Example: Fano's Projection ($n=2$)
 $\# \text{ points} = 7 = n^2 + n + 1$



c) The total number of lines in a projection plane is $n^2 + n + 1$:

Image to left shows $n=3$
and total lines at 13.

4.



"Hesse
Configuration

a) Each point on Affine plane has $n+1$ lines through it:

Example: Hesse Configuration

$$\text{Lines} = 4 = n+1$$

$n = 3$ points on each line

b) The total number of points in A is n^2 :

Example: Hesse Configuration

$$\text{Total points} = n^2$$

$$= 9, \text{ where } n=3.$$

c) The total number of lines in affine planes are $n(n+1)$:

Example: Hesse Configuration

$$\text{Total lines} = n(n+1)$$

$$= 12, \text{ where } n=3$$

5.

 F^2 is isomorphic:

Requirement: 1) (pg 69) "Incidence Axiom 3"

"There exists three distinct points with the property that no line is incident with all three of them"

$$\text{If } F^2 = \{(0,0), (0,1), (1,0), (1,1)\}$$

$$\text{then } y=0, y=1, x=0, x=1, y=x$$

and $y=-x+1$ are not

incident to three points.

2) (pg 69) "Incidence Axiom 3"

"For every line l , there exists at least two points with l ."

If a line is $ax+by+c=0$,

then

$a \neq 0, b \neq 0$	$a = 0$	$b = 0$	$a = 0, b = 0$
$(0,0), (1,1)$	$(0, -c/b)$	$(-c/a, 0)$	$(0,0), (0,1)$
$(0,1), (1,0)$	$(1, -c/b)$	$(-c/a, 1)$	$(1,0), (1,1)$

are combos to two points.

3) (pg 69) "Incidence Axiom 1"

"For every point P and for every point Q not equal to P , there exists a unique line l incident with P and Q ."

Line	$x=0$	$y=0$	$y=x$	$y=-x+1$	$x=1$	$y=1$
Points	$(0,0)$ $(0,1)$	$(0,0)$ $(1,0)$	$(0,0)$ $(1,1)$	$(1,0)$ $(0,1)$	$(1,0)$ $(1,1)$	$(0,1)$ $(1,1)$

4) (pg 21) "Euclidean Parallel Property"

"For every line l and every point P that does not lie on l , there exists a unique line m through P that is parallel to l ."

Line	$y=0$	$x=0$	$y=-x+1$	$y=x$
Parallel	$y=1$	$x=1$	$y=-x+2$	$y=x-1$

$P^2(F)$ is Isomorphic:

Requirements: 1) One-to-one Points.

F^2	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$P(F^2)$	$(0,0,1)$	$(0,1,1)$	$(1,0,1)$	$(1,1,1)$

2) one-to-one
Lines

$$F^2 = \{(0,0) | ax+bx+c=0\}$$

$$P(F^2) = \{(0,0,1) | ax+bx+c=0\}$$

$$a=b=c=0$$

$$F^2 = \{(0,1) | ax+by+c=0\}$$

$$P(F^2) = \{(0,1,1) | ax+by+c=0\}$$

$$a=0, b=-c$$

$$F^2 = \{(1,0) | ax+by+c=0\}$$

$$P(F^2) = \{(1,0,1) | ax+by+c=0\}$$

$$a=-c, b=0$$

$$F^2 = \{(1,1) | ax+by+c=0\}$$

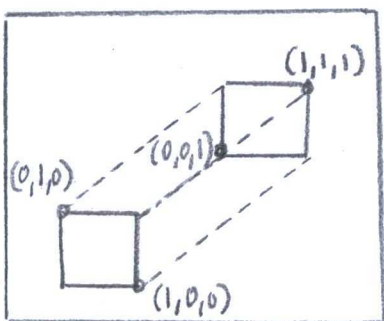
$$P(F^2) = \{(1,1,1) | ax+by+c=0\}$$

$$y = -\frac{a}{b}x - c$$

Map a line at infinity in $P(F^2)$:

$ax+by+c=0$ maps to $z=0$ by $a=0$
 $b=0$
and
 $c=1$.

6.



(from major exercise) "Fanos Axiom"

"For projective planes assert that the diagonal points of any quadrangle are not collinear"

Fanos plane is not a quadrangle
because three sides intersection
to a triangle.

$P^2(F)$:

Vertices: $(1,0,0), (0,1,0), (0,0,1), (1,1,1)$

Sides: #1) $a(x-1)=0$

#2) $b(y-1)=0$

#3) $c(z-1)=0$

#4) $aX=0$

#5) $bY=0$

#6) $cZ=0$

opposite: $1 \leftrightarrow 4$

Sides $2 \leftrightarrow 5$

$3 \leftrightarrow 6$

Diagonal: $(1,1,1), (0,0,1), (0,1,0), (1,1,1)$
Pairs

7.

Projective Planes requirement:

Other authors

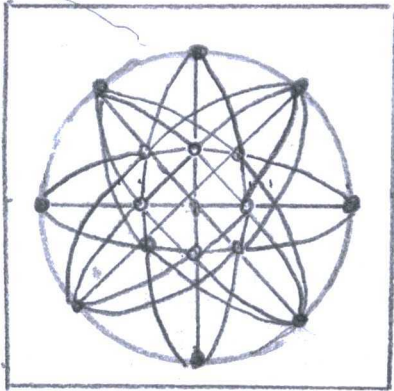
- Incidence Axiom #1
- Elliptic Parallel Property
- Existence of a Quadrangle

Marvin Jay Greenberg:

- Incidence Axiom #2
- Elliptic Parallel Property

Other authors described Incidence Axiom #1 as a critical to Projective Planes. Rather, M. Greenberg implied Incidence Axiom #2.

8.



"Projective Plane of order 3"

$$F = \{1, 1, -1\}$$

Addition:

	0	1	-1
0	0	1	-1
-1	1	-1	0
1	1	0	1

Multiplication:

	0	1	-1
0	0	0	0
-1	0	1	-1
1	0	-1	1

Labels to Figure:

$$\begin{array}{ccccc}
 (1, -1) & (0, -1) & & & (-1, -1) \\
 & (-1, 0) & (0, 1) & (1, 1) & \\
 (1, 0) & (-1, 0) & (0, 0) & (1, 0) & (-1, 0) \\
 & (-1, -1) & (0, -1) & (1, -1) & \\
 (1, 1) & (0, 1) & & & (-1, 1)
 \end{array}$$